Egypt-Focused Conversational Car Recommendation System

Objective

To build an AI-driven conversational assistant that helps car buyers in Egypt choose the car that best fits their budget, needs, and lifestyle. The system combines a data-driven recommendation engine with a natural-language chatbot (Arabic and English) that understands the buyer's requirements, recommends suitable cars, explains its choices, and refines its suggestions through conversation until the buyer finds the car they prefer.

Problem Statement & Motivation

Choosing a suitable car can be a complicated process, particularly for individuals who are unfamiliar with the automotive market. Potential buyers often face several challenges:

1. **Information Overload:** The automotive market offers numerous brands, models, and specifications, making it difficult to identify the most suitable options.
2. **Unclear Requirements:** Many buyers know their budget and general needs but may not understand which technical specifications or vehicle features are important for their intended usage.
3. **Time-Consuming Research:** Buyers often need to visit multiple websites, compare specifications, and manually evaluate different vehicles.
4. **Lack of Personalized Guidance:** Traditional car-search platforms mainly depend on predefined filters and do not provide interactive guidance that helps users refine their choices.

The proposed platform addresses these challenges through an intelligent conversational recommendation system that understands users' requirements, analyzes available vehicle data, and provides personalized, explainable recommendations.

Project Description

This project builds a platform that extracts the buyer's preferences from a free-form conversation (for example: budget, family size, daily distance, city or highway driving, new or used), ranks real cars from a local database, and explains each recommendation. The system will be designed so the data source is replaceable: it will start with public market data and can later be connected to the inventory of a car dealership.

The project combines software development, data analytics, and AI. Its recommendations will be evaluated against baselines (filter-based search and a general-purpose chatbot) using measurable metrics and user testing.

Key Features include:

1. **Conversational Preference Gathering:** A chatbot that asks about budget, usage, and preferences, and converts free text (Arabic or English) into structured requirements.
2. **Personalized Car Recommendation Engine:** Multi-criteria ranking of cars from the database based on the buyer's requirements. Buyers can refine the results during the conversation (for example: "cheaper", "more spacious", "lower fuel consumption").
3. **Explainable Recommendations & Car Comparison:**
    - Displays recommended vehicles with their specifications and estimated prices.
    - Explains why each vehicle matches the user's requirements.
    - Highlights advantages and disadvantages based on the user's priorities.
    - Allows users to compare multiple recommended vehicles side by side.
4. **Grounded Answers (RAG):** The chatbot answers questions about specs, prices, and comparisons using retrieved data from the car database, to avoid made-up information.
5. **Market Analytics Dashboard:** Visual insights on prices, popular models, depreciation, and price ranges by brand, year, and body type.
6. **Dealership Mode (optional):** A configurable mode that limits recommendations to a specific dealership's inventory.

Technologies to Use

- **Frontend:** React (chat interface, comparison pages, dashboard)
- **Backend:** Python (FastAPI) with REST APIs
- **Database:** SQL (PostgreSQL or SQLite) for cars, users, and conversations
- **Machine Learning:**
    - Recommender System: Scikit-learn (content-based filtering, multi-criteria ranking)
    - NLP: Transformers (multilingual sentence embeddings for Arabic/English understanding and preference extraction)
    - Conversational AI: Large Language Model (hosted API or open-source model) with Retrieval-Augmented Generation (RAG)
    - Vector Search: FAISS or ChromaDB
    - Price Analysis (optional): regression models for price estimation
- **Data Processing:** Pandas, NumPy
- **Data Visualization:** Plotly
- **Data Sources:** Public car listing data, manufacturer and agent specifications, and open datasets (collected for academic use and in line with each source's terms)

Weekly Milestones & Deliverables

Week 1: Planning, Data Collection & Setup

- Define the project scope, target users, and the car categories to cover.
- Design the system architecture and the database schema for cars, users, and conversations.
- Collect the initial car dataset from public sources and clean it (remove duplicates, fix missing values and inconsistent names).
- Set up the development environment, version control, and project structure.

Week 2: Recommendation Engine & Application Core

- Implement the multi-criteria recommendation model, together with a simple filter-based search to use as a baseline.
- Build the backend APIs and the basic web application (chat screen and car listing).
- Begin the market analysis of prices and trends for the dashboard.

Week 3: Conversational Layer & Explainable Recommendations

- Implement preference extraction from free-text conversation in Arabic and English, and the dialogue flow.
- Implement retrieval-augmented generation (RAG) so answers are grounded in the car database.
- Build the explainable recommendation view and the side-by-side comparison page.
- Complete the first version of the analytics dashboard.

Week 4: Evaluation, Refinement & Final Presentation

- Compare the system against the baselines (filter-based search and a general chatbot) using relevance metrics.
- Run user testing with a group of participants and collect feedback through a questionnaire.
- Fix bugs, improve recommendations and conversation quality, and finalize the optional dealership mode.
- Prepare the final report, documentation, and presentation.